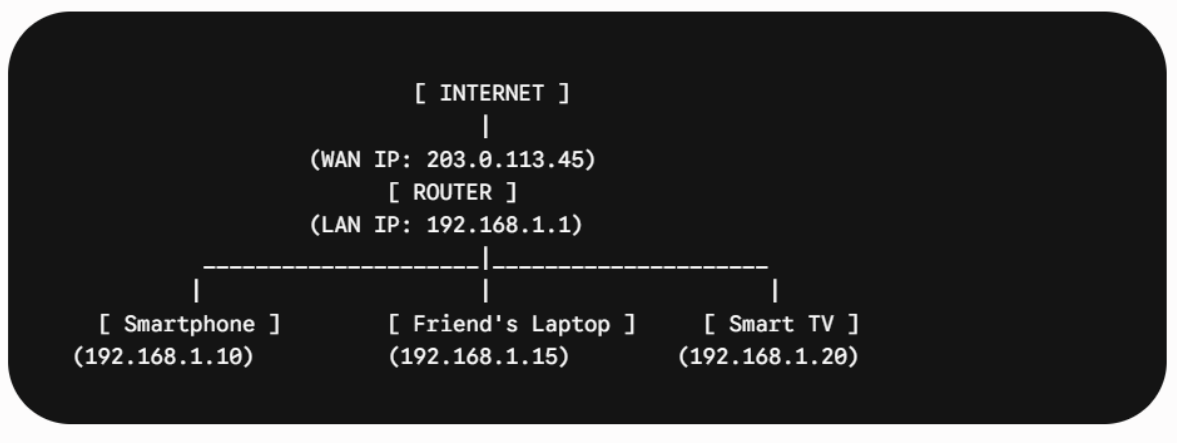


1. Network Diagram Representation



IP Address Assignments

- **Router WAN Interface (Public IP):** 203.0.113.45 (*facing the ISP/Internet*)
- **Router LAN Interface (Gateway IP):** 192.168.1.1 (*facing the local network*)
- **Smartphone:** 192.168.1.10
- **Friend's Laptop:** 192.168.1.15
- **Smart TV:** 192.168.1.20
- **Subnet Mask for LAN:** 255.255.255.0 (/24)

2: Step-by-Step Manual IP Routing Simulation

Network Setup

- **Your Laptop (Source):** 192.168.1.50/24 (Gateway: 192.168.1.1)
- **Friend's Laptop (Destination):** 10.0.2.75/24
- **Router A (Local Gateway):**
 - Interface eth0 (LAN): 192.168.1.1/24
 - Interface eth1 (WAN link to Router B): 172.16.0.1/30
- **Router B (Remote Gateway):**
 - Interface eth0 (WAN link to Router A): 172.16.0.2/30
 - Interface eth1 (Remote LAN): 10.0.2.1/24

Step 1: Source Host (Your Laptop)

Your laptop wants to send data to 10.0.2.75.

1. **Subnet Check:** The laptop compares 10.0.2.75 against its own IP and subnet (192.168.1.50/24). Because 10.0.2.75 is in a different subnet, it cannot deliver the packet directly using MAC addresses.
2. **Action:** The laptop sends the IP packet to its default gateway (192.168.1.1).

Step 2: First Hop (Router A)

Packet Header Received:

- **Source IP:** 192.168.1.50
- **Destination IP:** 10.0.2.75

Router A inspects its routing table:

Destination Network	Subnet Mask	Next Hop	Interface
192.168.1.0	255.255.255.0	Directly Connected	eth0
172.16.0.0	255.255.255.252	Directly Connected	eth1
10.0.2.0	255.255.255.0	172.16.0.2	eth1
0.0.0.0	0.0.0.0	172.16.0.2	eth1

Routing Decision: Match found for 10.0.2.0/24. The next hop is 172.16.0.2 via eth1. Router A decrements the TTL (Time-To-Live) by 1 and forwards the packet out of interface eth1.

Step 3: Second Hop (Router B)

Packet Header Received:

- **Source IP:** 192.168.1.50
- **Destination IP:** 10.0.2.75

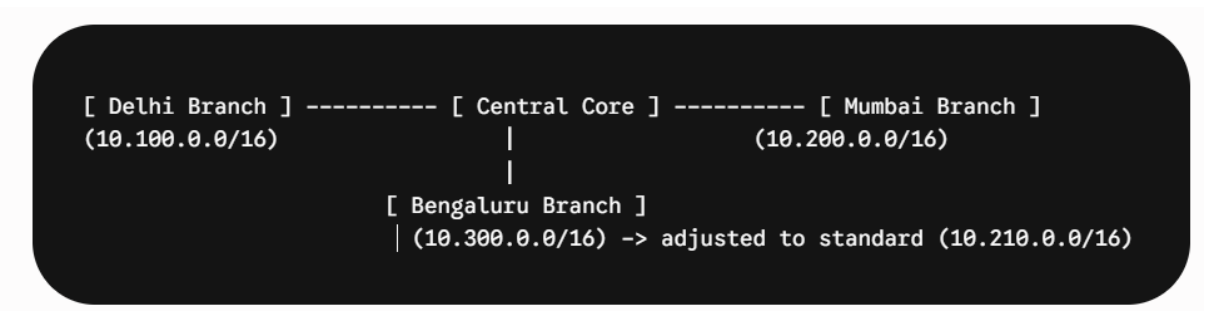
Router B inspects its routing table:

Destination Network	Subnet Mask	Next Hop	Interface
172.16.0.0	255.255.255.252	Directly Connected	eth0
10.0.2.0	255.255.255.0	Directly Connected	eth1

Routing Decision: Match found for 10.0.2.0/24. This network is **directly connected** to interface eth1. Router B resolves 10.0.2.75's MAC address using ARP and delivers the packet directly to your friend's laptop.

Task 3: Zomato-Style Delivery Network

Scenario: A regional food delivery platform connects three main hub branches (Delhi, Mumbai, Bengaluru) through a central backbone core (172.20.0.0/30 links).



Below is the consolidated routing table for the **Central Core Router**:

Branch / Destination	Network Address	Subnet Mask	Next Hop IP	Outgoing Interface	Note / Purpose
Delhi Hub	10.100.0.0	255.255.0.0 (/16)	172.20.0.2	eth1	Kitchen & Fleet Tracking
Mumbai Hub	10.200.0.0	255.255.0.0 (/16)	172.20.0.6	eth2	Order Dispatch & POS
Bengaluru Hub	10.210.0.0	255.255.0.0 (/16)	172.20.0.10	eth3	Tech HQ & Local Servers
Internet Gateway	0.0.0.0	0.0.0.0 (/0)	203.0.113.1	wan0	Default Route (Customer App API)

Task 4: Static Routing vs. Dynamic Routing

Comparison Matrix

Aspect	Static Routing	Dynamic Routing (e.g., OSPF, RIP)
Route Selection	Manually configured by engineers	Automatically calculated by dynamic algorithms
Adaptability	Rigid; fails if a link goes down	Dynamic; automatically reroutes around network failures

Aspect	Static Routing	Dynamic Routing (e.g., OSPF, RIP)
Overhead	Very low CPU/RAM usage; zero bandwidth overhead	Requires extra CPU/RAM cycles and periodic routing updates

Pros & Cons

Static Routing

- **Pros:**
 1. **Predictable & Secure:** No routing updates are broadcasted, reducing the risk of route spoofing.
 2. **Zero Overhead:** Consumes no router memory or network bandwidth for exchanging tables.
 3. **Simple for Small Networks:** Extremely simple to configure for fixed setups like point-of-sale terminals.
- **Cons:**
 1. **No Automatic Failover:** If a primary link cuts out, traffic is dropped until an administrator manually fixes the route.
 2. **Unscalable:** Adding a new branch requires manually updating every router on the network.
 3. **Human Error Prone:** Typos in static routes can cause routing loops or dead ends.

Dynamic Routing (OSPF / RIP)

- **Pros:**
 1. **Automatic Failover & Rerouting:** Automatically redirects traffic around broken links within seconds.
 2. **High Scalability:** Adding a new router or city branch automatically populates routes across the entire network.
 3. **Optimal Path Selection:** Automatically calculates the fastest route based on metrics like link speed or hop count.
- **Cons:**
 1. **Resource Intensive:** Uses CPU, RAM, and network bandwidth to maintain topology maps.

2. **Complex Configuration:** Misconfigurations can lead to routing flaps or protocol instability.
3. **Security Risks:** Vulnerable to rogue routing updates if authentication isn't enforced.

Real-World Food Delivery Analogy (Zomato / Swiggy)

- **Static Routing is like fixed delivery zones:** A delivery driver is hardcoded to only take Route A from the restaurant to a delivery location. If road construction blocks Route A, the driver gets stuck and cannot deliver the order until someone tells them a new route.
- **Dynamic Routing is like Google Maps navigation for riders:** The driver constantly gets real-time traffic updates. If an accident occurs on the main highway, the app automatically recalculates a detour through side streets to make sure your food arrives hot.

Plaintext

```
! --- Router A Configuration ---
Router_A# configure terminal
Router_A(config)# router ospf 1
Router_A(config-router)# router-id 1.1.1.1
Router_A(config-router)# network 192.168.1.0 0.0.0.255 area 0
Router_A(config-router)# network 10.0.0.0 0.0.0.3 area 0
Router_A(config-router)# exit

! --- Router B Configuration ---
Router_B# configure terminal
Router_B(config)# router ospf 1
Router_B(config-router)# router-id 2.2.2.2
Router_B(config-router)# network 10.0.0.0 0.0.0.3 area 0
Router_B(config-router)# network 192.168.2.0 0.0.0.255 area 0
Router_B(config-router)# exit
```

Line-by-Line Explanation

1. **configure terminal**

- *What it does:* Enters global configuration mode so you can apply system-level commands to the router.

2. **router ospf 1**

- *What it does:* Initializes the OSPF routing process. The number 1 is the **Process ID** (a local identifier for this OSPF instance on the router).

3. **router-id 1.1.1.1** (and 2.2.2.2)

- *What it does:* Gives the router a unique 32-bit identifier in the OSPF network. This helps neighbor routers identify who is sending updates.

4. **network 192.168.1.0 0.0.0.255 area 0**

- *What it does:* Tells the router to enable OSPF on interfaces matching the 192.168.1.0/24 network range and assign them to **Area 0** (the backbone area).
- *Note:* 0.0.0.255 is a **wildcard mask** (the inverse of subnet mask 255.255.255.0).

5. **network 10.0.0.0 0.0.0.3 area 0**

- *What it does:* Enables OSPF on the point-to-point link connecting Router_A and Router_B (10.0.0.0/30). Once both routers match this network, they establish an OSPF adjacency and exchange routes automatically!